

## Module 3

# Measuring Corporate Performance

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FINE 3010, Financial Management  
Prof. Renping Li, Tulane University

# How Do We Evaluate a Corporation's Performance?

- Who needs to evaluate corporate performance?
  - Managers
  - Current and prospective equity investors
  - Current and prospective debt investors

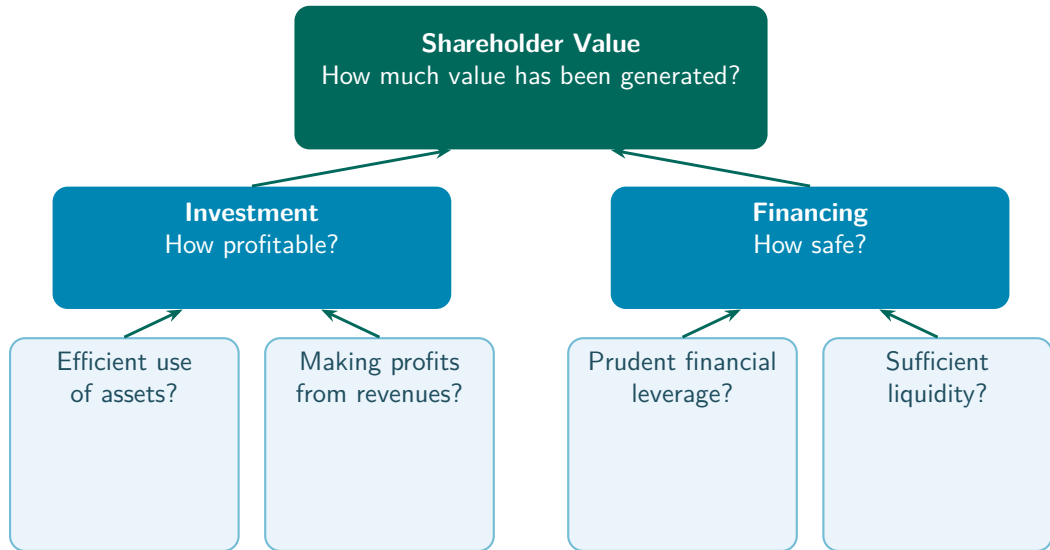


# How Do We Evaluate a Corporation's Performance?

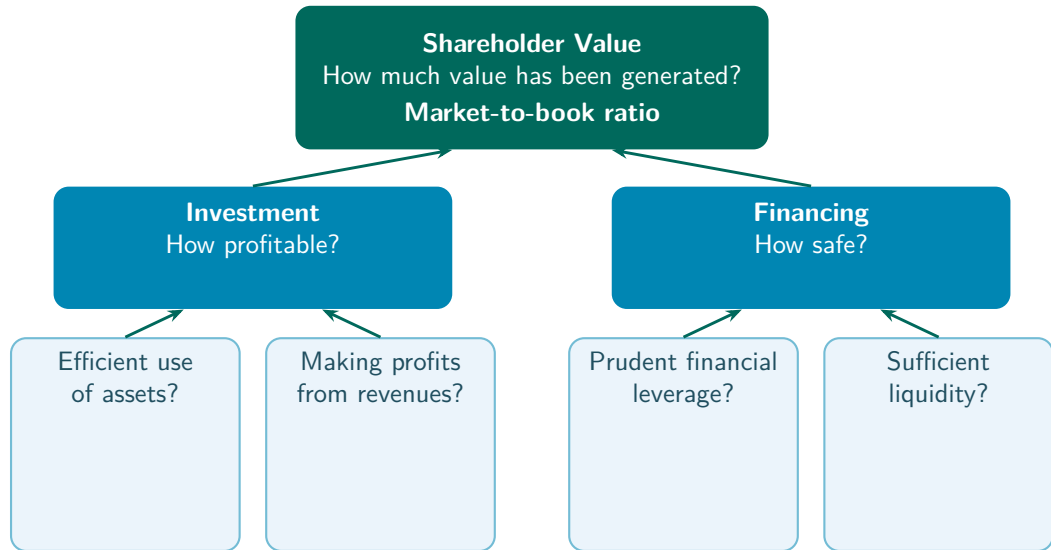
- Who needs to evaluate corporate performance?
  - Managers
  - Current and prospective equity investors
  - Current and prospective debt investors
- Corporate performance is **multi-dimensional**



# Measuring Corporate Performance



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# Shareholder Value

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- **Market capitalization**, another name for the market value of equity

## Example, Apple

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*“some kind of fruit company”*



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## Example, Apple

- Recall from **Module 2**, Apple's book value of equity at the end of fiscal year 2025 was \$73,733 Mn
- At the end of 2025, share price about \$271, shares outstanding 14,800 Mn
- Market capitalization =  $\$271 \times 14,800$  Mn = \$4,010,800 Mn



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# Shareholder Value, Market-to-Book Ratio

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## Pros

- Closely tied to how actual investors value the company

## Cons

- Subject to market fluctuations

## Practice Problem

- Here is a simplified balance sheet for a business (figures in \$ millions). It has 600 million shares outstanding with a market price of \$80 a share.

|                     |                 |                                     |                 |
|---------------------|-----------------|-------------------------------------|-----------------|
| Current assets      | \$40,000        | Current liabilities                 | \$30,000        |
| Fixed assets        | 50,000          | Long-term liabilities               | 44,000          |
|                     |                 | Shareholders' equity                | 16,000          |
| <b>Total assets</b> | <b>\$90,000</b> | <b>Total liabilities and equity</b> | <b>\$90,000</b> |

- Calculate the market-to-book ratio.

## Practice Problem

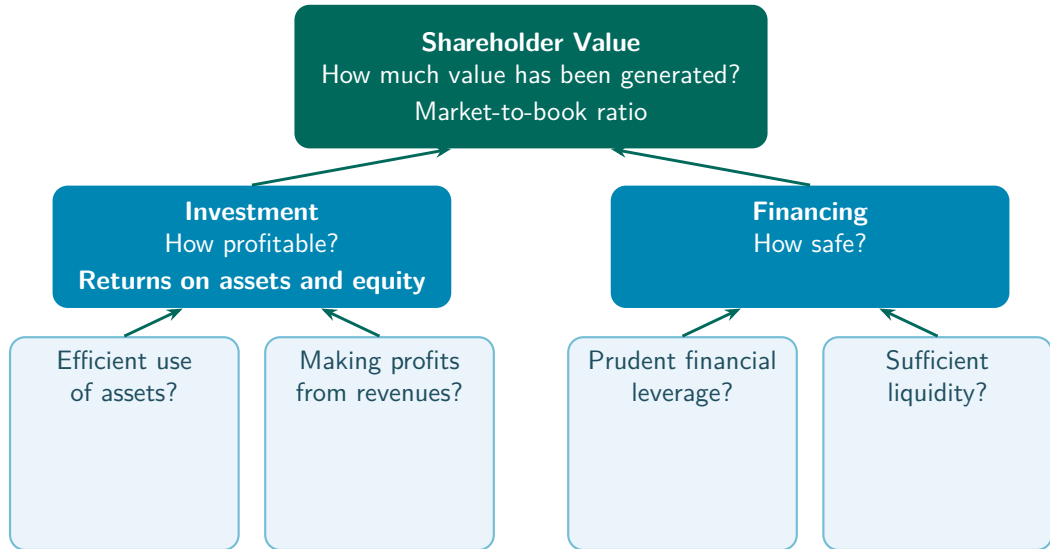
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- Calculate the market-to-book ratio.  $600 \times \$80 / \$16,000 = \$48,000 / \$16,000 = 3.0$



# Measuring Corporate Performance



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## Pros

- Less sensitive to stock market fluctuations

## Cons

- Book values may deviate from market values

## Example, Income Statement (\$ Millions)

|                               | \$ Million   |
|-------------------------------|--------------|
| Revenues                      | 80,000       |
| COGS                          | 56,000       |
| SG&A expenses                 | 16,000       |
| Depreciation                  | 3,000        |
| <b>EBIT</b>                   | <b>5,000</b> |
| Interest expense              | 1,000        |
| <b>Taxable income</b>         | <b>4,000</b> |
| Taxes                         | 840          |
| <b>Net income</b>             | <b>3,160</b> |
| Allocation of net income      |              |
| Dividends                     | 1,000        |
| Addition to retained earnings | 2,160        |

## Example, Balance Sheet (\$ Millions)

| Assets                     |               | Liabilities and equity              |               |
|----------------------------|---------------|-------------------------------------|---------------|
| Current assets             |               | Current liabilities                 |               |
| Cash and marketable sec.   | 2,600         | Debt due for repayment              | 200           |
| Accounts receivable        | 500           | Accounts payable                    | 9,900         |
| Inventories                | 9,000         | Total current liabilities           | 10,100        |
| Total current assets       | 12,100        | Long-term liabilities               | 11,300        |
| Fixed assets               |               | Total liabilities                   | 21,400        |
| Tangible fixed assets      |               | Shareholders' equity                |               |
| Property, plant, equipment | 48,200        | Paid-in capital                     | 6,300         |
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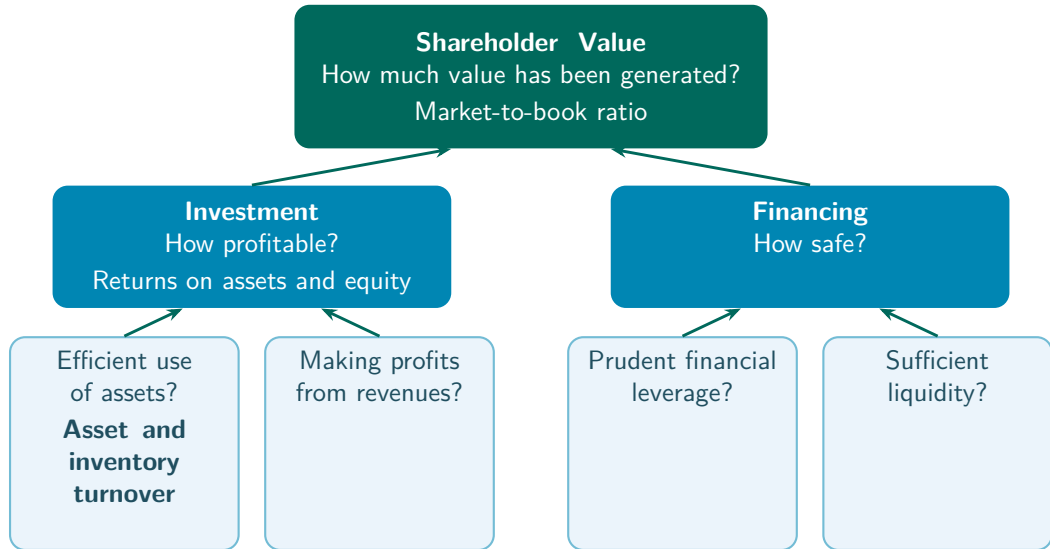
## Practice Problem

- Last year's ROE was 30%. This year the ROE has decreased to 20%, even though the firm's net income equaled last year's. What caused the decrease?
  - a. Shareholders' equity decreased by 10%.
  - b. Shareholders' equity decreased by 50%.
  - c. Shareholders' equity increased by 10%.
  - d. Shareholders' equity increased by 50%.

## Practice Problem

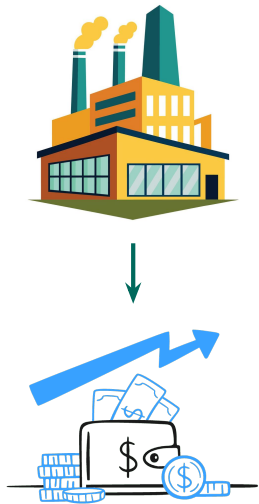
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  - d. Shareholders' equity increased by 50%.
- Suppose net income is  $X$  in both years. Then  $30\% = X / \text{shareholders' equity last year}$ , and  $20\% = X / \text{shareholders' equity this year}$
- Then  $X = 30\% \times \text{shareholders' equity last year} = 20\% \times \text{shareholders' equity this year}$
- So  $\text{shareholders' equity this year} / \text{shareholders' equity last year} = 30\%/20\% = 150\%$

# Measuring Corporate Performance



# Efficiency, Use of Assets

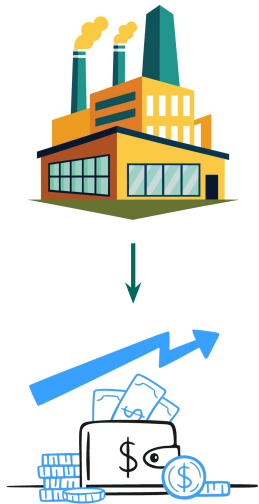
$$\text{Asset turnover ratio} = \frac{\text{Revenues}}{\text{Total assets}}$$



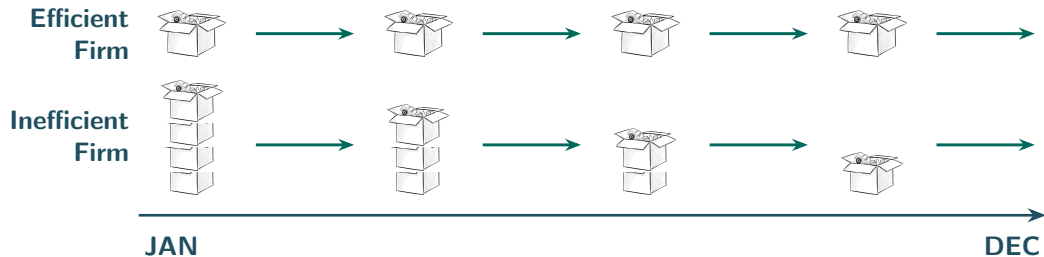
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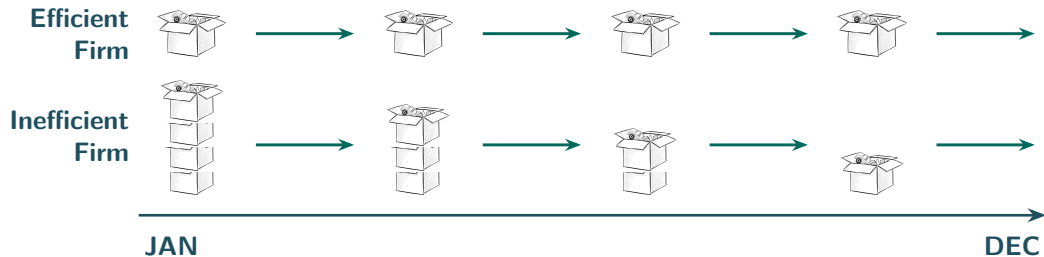
- How much in revenues is generated by each dollar of assets?



# Efficiency, Inventory

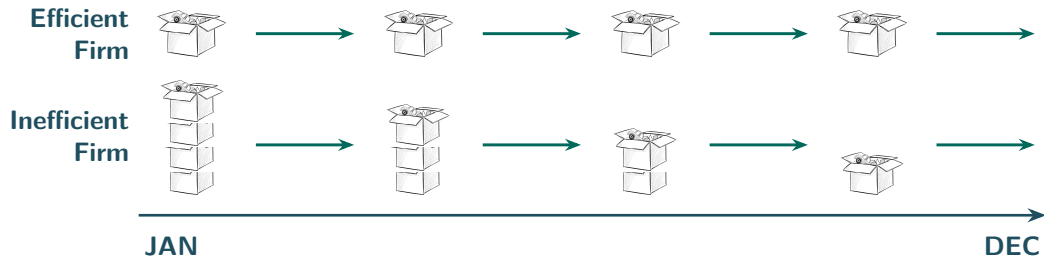


# Efficiency, Inventory



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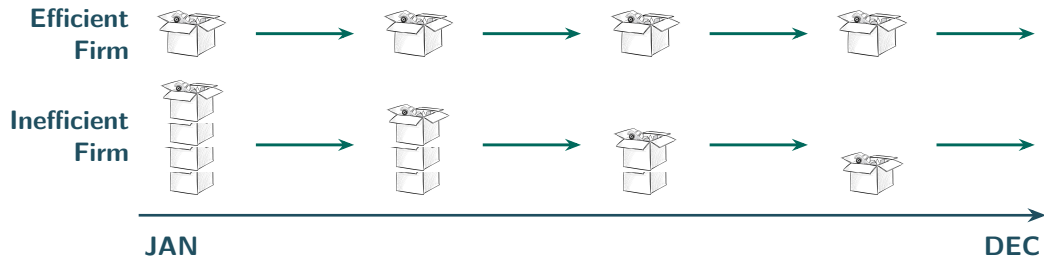


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- Efficient firms do not let inventories sit idle
- Aim for the highest possible? No, inadequate inventories may fail to meet customer demand in time

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- The average length of time for customers to pay their bills
- Efficient firms collect bills quickly
- Aim for the fastest possible? No, it may reflect an over-stringent credit policy
  - **Credit policy**, the terms on which a firm lets customers buy now and pay later

## Example, Income Statement (\$ Millions)

|                               | \$ Million   |
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| Revenues                      | 80,000       |
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$$\text{Average collection period} = \frac{500}{80,000/365} = 2.3 \text{ days}$$

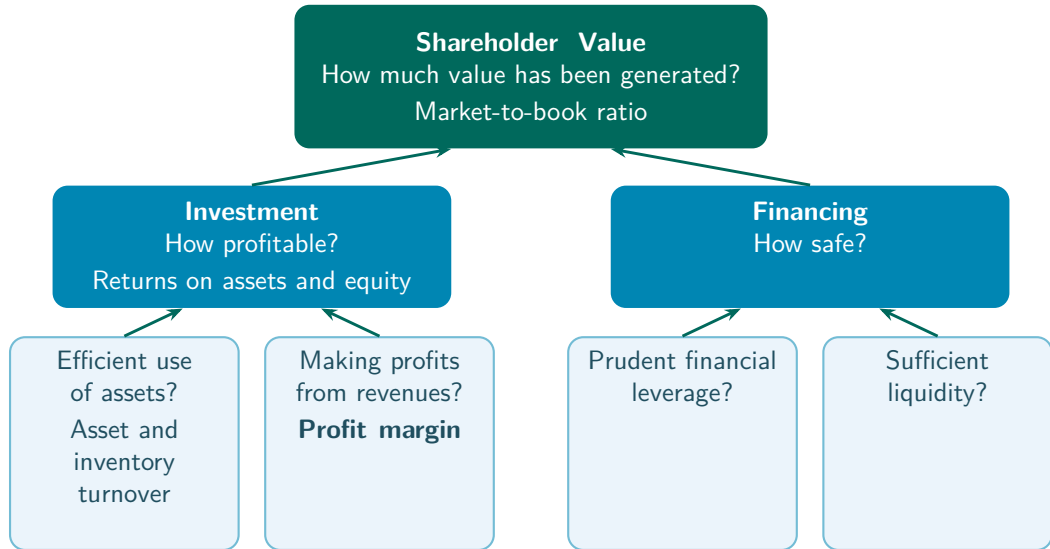
## Practice Problem

- A firm had accounts receivable of \$940 at the beginning of the year and generated \$3,400 of revenues during the year. How much is its average collection period?
  - a. 94 days
  - b. 95 days
  - c. 101 days
  - d. 107 days

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  - d. 107 days
- Average collection period =  $940 / (3,400 / 365) = 101$  days

# Measuring Corporate Performance



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$$\text{Profit margin} = \frac{\text{Net income}}{\text{Revenues}}$$

- How much of each dollar of revenues ends up as net income?

# Discuss, AI and Profit Margins

- Will AI raise or lower firm profit margins overall?

**Aswath Damodaran**, “Dean of Valuation”



Excess Returns, January 22, 2026  
(1:23), “Could AI Actually Reduce  
Profit Margins? Aswath Damodaran  
Thinks So”

# Discuss, AI and Profit Margins

- Will AI raise or lower firm profit margins overall?
- Do you agree?

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$$\text{ROA} = \frac{\text{Revenues}}{\text{Assets}} \times \frac{\text{Net income}}{\text{Revenues}}$$



# Decompose ROA

- We can separate ROA into component ratios

$$\text{ROA} = \frac{\text{Net income}}{\text{Assets}}$$

$$\text{ROA} = \frac{\text{Revenues}}{\text{Assets}} \times \frac{\text{Net income}}{\text{Revenues}}$$

$$\text{ROA} = \text{Asset turnover} \times \text{Profit margin}$$

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$$\text{ROA} = \frac{\text{Revenues}}{\text{Assets}} \times \frac{\text{Net income}}{\text{Revenues}} = \text{Asset turnover} \times \text{Profit margin} = 1.94 \times 0.0395 = 0.077$$

# Decompose ROA

| Company | Strategy       | Asset Turnover | Profit Margin | ROA |
|---------|----------------|----------------|---------------|-----|
| Zara    | Fast fashion   | 1.6            | 7.5%          | 12% |
| Hermès  | Luxury fashion | 0.55           | 22.5%         | 12% |



## Practice Problem

- Which of the following will allow your firm to achieve its targeted 16% ROA with an asset turnover of 2.5?
  - a. A current ratio of 1.6.
  - b. A P/E ratio of 14.
  - c. A return on equity of 25%.
  - d. A profit margin of 6.4%.

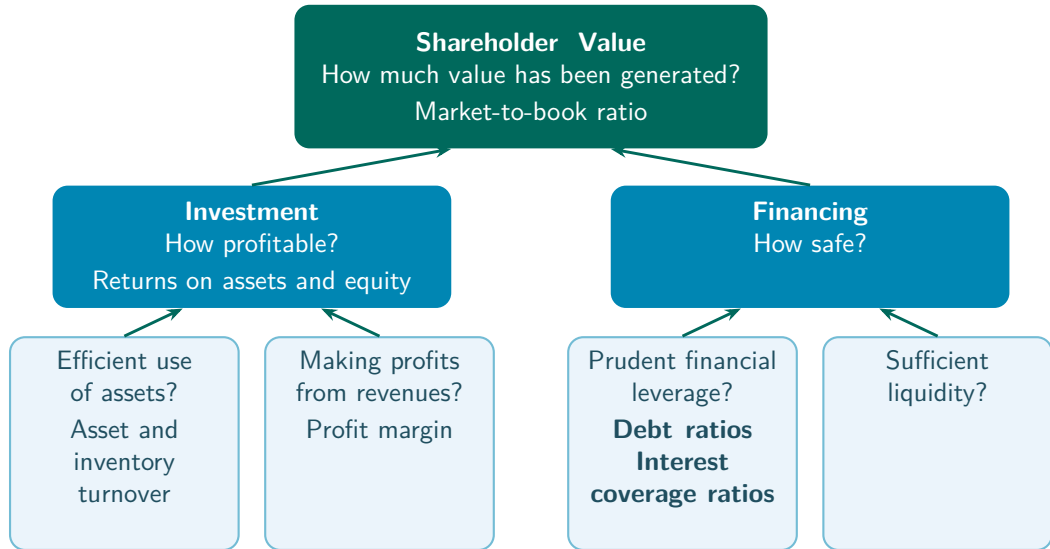
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- $ROA = \text{Asset turnover} \times \text{Profit margin}$ , and  $2.5 \times 6.4\% = 16\%$

## Recap, Investment

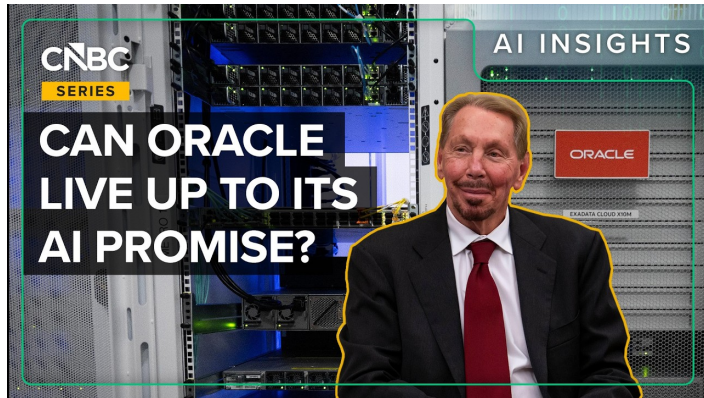
- Investment profitability is measured with the returns on assets and equity
- Turnover ratios, how much in revenues the assets generate and how quickly inventories sell
- Profit margin, how much of each dollar of revenues ends up as net income
- $ROA = \text{Asset turnover} \times \text{Profit margin}$

# Measuring Corporate Performance





# Motivating Example, Oracle



CNBC, March 10, 2026 (5:45), “How Oracle’s AI-Fueled Debt Load Has Investors On Edge”

## Example, Financing a Data Center

- Suppose building a data center costs \$2B, and it will generate \$6B (**good**, 50% chance) or \$2B (**bad**, 50% chance)

## Example, Financing a Data Center

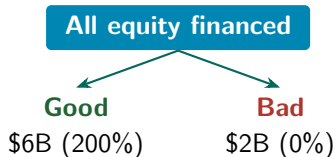
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## Example, Financing a Data Center

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- Assume the interest rate on debt is 50%

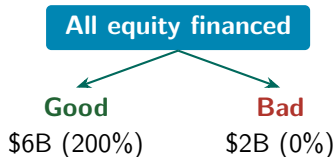
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**\$1B equity + \$1B debt**

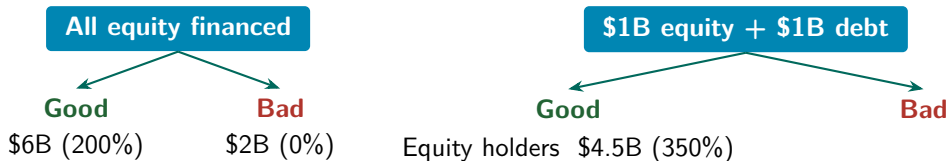
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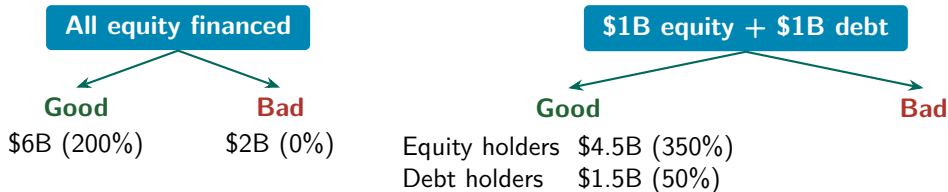
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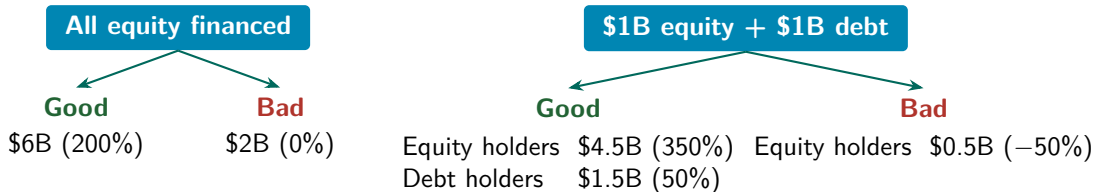
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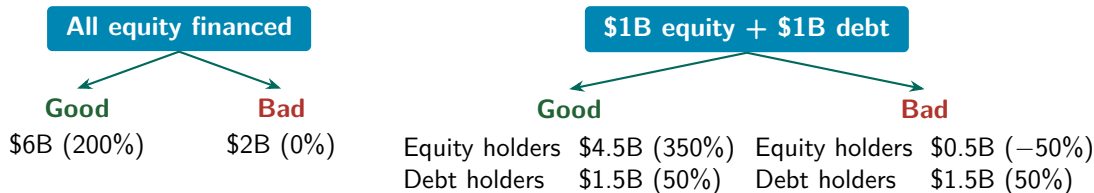
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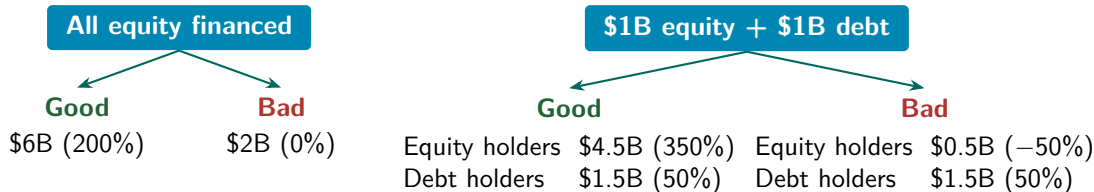
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**With debt, gains and losses to equity holders are amplified**

# Financial Leverage

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- Expansion fails and profits drop → debt holders are paid first, shareholders may be left with nothing
- Debt increases returns to shareholders in good times and reduces them in bad times





# Financial Leverage

- Suppose a firm borrows money to invest in real assets and expand
- Expansion succeeds and profits rise → debt holders still receive only the fixed interest and principal, the gains go to shareholders
- Expansion fails and profits drop → debt holders are paid first, shareholders may be left with nothing
- Debt increases returns to shareholders in good times and reduces them in bad times
  - This is **financial leverage**



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$$\text{Total debt ratio} = \frac{\text{Total liabilities}}{\text{Total assets}}$$

## Example, Balance Sheet (\$ Millions)

| Assets                     |               | Liabilities and equity              |               |
|----------------------------|---------------|-------------------------------------|---------------|
| Current assets             |               | Current liabilities                 |               |
| Cash and marketable sec.   | 2,600         | Debt due for repayment              | 200           |
| Accounts receivable        | 500           | Accounts payable                    | 9,900         |
| Inventories                | 9,000         | Total current liabilities           | 10,100        |
| Total current assets       | 12,100        | Long-term liabilities               | 11,300        |
| Fixed assets               |               | Total liabilities                   | 21,400        |
| Tangible fixed assets      |               | Shareholders' equity                |               |
| Property, plant, equipment | 48,200        | Paid-in capital                     | 6,300         |
| Less accum. depreciation   | 19,700        | Retained earnings                   | 13,600        |
| Net tangible fixed assets  | 28,500        | Total shareholders' equity          | 19,900        |
| Intangible asset           | 700           |                                     |               |
| Total fixed assets         | 29,200        |                                     |               |
| <b>Total assets</b>        | <b>41,300</b> | <b>Total liabilities and equity</b> | <b>41,300</b> |

## Example, Leverage

$$\text{Long-term debt ratio} = \frac{11,300}{11,300 + 19,900} = 0.36$$

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$$\text{Long-term debt-to-equity ratio} = \frac{11,300}{19,900} = 0.57$$

$$\text{Total debt ratio} = \frac{\text{Total liabilities}}{\text{Total assets}} = \frac{21,400}{41,300} = 0.52$$



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## Example, Income Statement (\$ Millions)

|                               | \$ Million   |
|-------------------------------|--------------|
| Revenues                      | 80,000       |
| COGS                          | 56,000       |
| SG&A expenses                 | 16,000       |
| Depreciation                  | 3,000        |
| <b>EBIT</b>                   | <b>5,000</b> |
| Interest expense              | 1,000        |
| <b>Taxable income</b>         | <b>4,000</b> |
| Taxes                         | 840          |
| <b>Net income</b>             | <b>3,160</b> |
| Allocation of net income      |              |
| Dividends                     | 1,000        |
| Addition to retained earnings | 2,160        |

## Example, Leverage

$$\text{Times interest earned} = \frac{\text{EBIT}}{\text{Interest payments}} = \frac{5,000}{1,000} = 5.0$$

## Example, Leverage

$$\text{Times interest earned} = \frac{\text{EBIT}}{\text{Interest payments}} = \frac{5,000}{1,000} = 5.0$$

$$\text{Cash coverage ratio} = \frac{\text{EBIT} + \text{Depreciation}}{\text{Interest payments}} = \frac{5,000 + 3,000}{1,000} = 8.0$$

## Practice Problem

- A firm has a times interest earned ratio of 2, a tax liability of \$1 million, and interest expense of \$1.5 million. Its book value of equity is \$1.5 million. How much is its ROE?
- a. 26.67%
  - b. 30.00%
  - c. 33.33%
  - d. 50.00%

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  - a. 26.67%
  - b. 30.00%
  - c. 33.33%
  - d. 50.00%
- Times interest earned =  $\text{EBIT} / \text{Interest expense} = 2$ , and interest expense = 1.5, so  $\text{EBIT} = 3$
- Net income =  $\text{EBIT} - \text{interest expense} - \text{taxes} = 3 - 1.5 - 1 = 0.5$
- $\text{ROE} = \text{Net income} / \text{Equity} = 0.5 / 1.5 = 33.33\%$

## Practice Problem

- A firm pays an 8% rate of interest on \$10 million of outstanding debt with face value \$10 million. The firm's EBIT was \$1 million.
  - a. What is its times interest earned?
  - b. If depreciation is \$200,000, what is its cash coverage ratio?



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Interest expense =  $0.08 \times \$10 \text{ million} = \$800,000$   
Times interest earned =  $\$1,000,000 / \$800,000 = 1.25$
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Interest expense =  $0.08 \times \$10 \text{ million} = \$800,000$   
Times interest earned =  $\$1,000,000 / \$800,000 = 1.25$
  - b. If depreciation is \$200,000, what is its cash coverage ratio?  
Cash coverage ratio =  $(\$1,000,000 + \$200,000) / \$800,000 = 1.5$

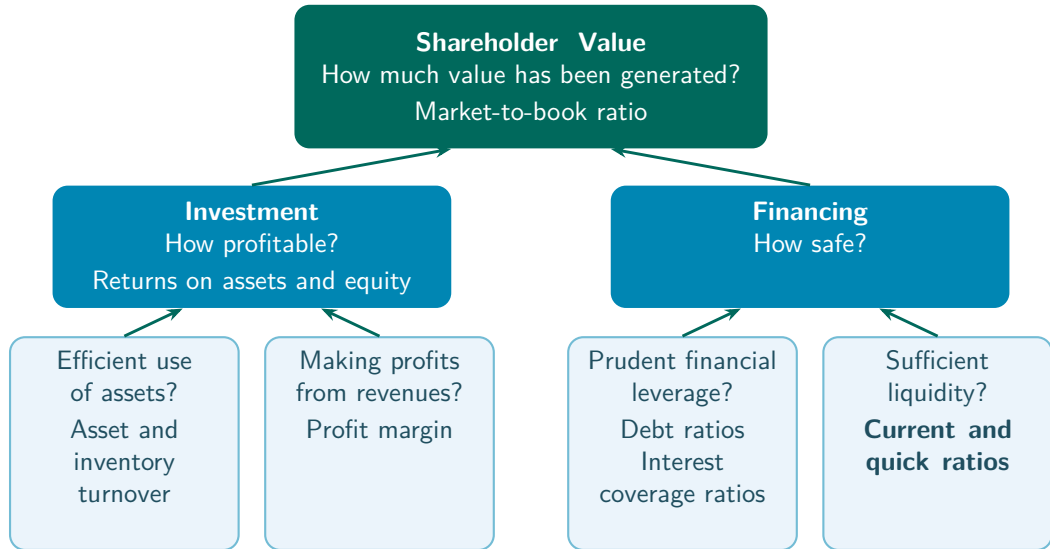
## Practice Problem

- Which one of the following will increase a firm's times interest earned ratio?
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  - b. A decrease in cost of goods sold.
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# Measuring Corporate Performance



# Financial Liquidity

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# Financial Liquidity

- **Liquidity**, the ability to sell an asset on short notice at close to the market value
- Liquid assets can be converted into cash quickly and cheaply
- Liquidity metrics compare the amount of liquid assets to the amount of debt that requires immediate repayment





## Example, Balance Sheet (\$ Millions)

| Most liquid                                                                       | Assets                     |               | Liabilities and equity              |               | Paid first                                                                          |
|-----------------------------------------------------------------------------------|----------------------------|---------------|-------------------------------------|---------------|-------------------------------------------------------------------------------------|
|                                                                                   |                            |               |                                     |               |                                                                                     |
|  | Current assets             |               | Current liabilities                 |               |  |
|                                                                                   | Cash and marketable sec.   | 2,600         | Debt due for repayment              | 200           |                                                                                     |
|                                                                                   | Accounts receivable        | 500           | Accounts payable                    | 9,900         |                                                                                     |
|                                                                                   | Inventories                | 9,000         | Total current liabilities           | 10,100        |                                                                                     |
|                                                                                   | Total current assets       | 12,100        | Long-term liabilities               | 11,300        |                                                                                     |
|                                                                                   | Fixed assets               |               | Total liabilities                   | 21,400        |                                                                                     |
|                                                                                   | Tangible fixed assets      |               | Shareholders' equity                |               |                                                                                     |
|                                                                                   | Property, plant, equipment | 48,200        | Paid-in capital                     | 6,300         |                                                                                     |
|                                                                                   | Less accum. depreciation   | 19,700        | Retained earnings                   | 13,600        |                                                                                     |
|                                                                                   | Net tangible fixed assets  | 28,500        | Total shareholders' equity          | 19,900        |                                                                                     |
| Least liquid                                                                      | Intangible asset           | 700           |                                     |               | Paid last                                                                           |
|                                                                                   | Total fixed assets         | 29,200        |                                     |               |                                                                                     |
|                                                                                   | <b>Total assets</b>        | <b>41,300</b> | <b>Total liabilities and equity</b> | <b>41,300</b> |                                                                                     |

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$$\frac{\text{Cash} + \text{Marketable securities} + \text{Accounts receivable}}{\text{Current liabilities}} = \frac{2,600 + 500}{10,100} = 0.31$$

## Practice Problem

- Consider this simplified balance sheet for a business.

| Balance Sheet (\$) |              |                     |              |
|--------------------|--------------|---------------------|--------------|
| Current assets     | \$100        | Current liabilities | \$60         |
| Long-term assets   | 500          | Long-term debt      | 280          |
|                    |              | Other liabilities   | 70           |
|                    |              | Equity              | 190          |
| <b>Total</b>       | <b>\$600</b> | <b>Total</b>        | <b>\$600</b> |

- What is the company's debt-to-equity ratio?
- What is the ratio of total long-term debt to total long-term capital?
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 $\$280/(\$280 + 190) = 0.60$
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- What is the ratio of total long-term debt to total long-term capital?  
 $\$280/(\$280 + 190) = 0.60$
- What is its current ratio?  $\$100/\$60 = 1.67$

## Practice Problem

- A firm uses \$1 million in cash to purchase inventories.
  - a. Will its current ratio rise or fall?
  - b. Will its quick ratio rise or fall?

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## Practice Problem

- A firm uses \$1 million in cash to purchase inventories.
  - a. Will its current ratio rise or fall?  
The current ratio is unaffected. Inventories replace cash, but total current assets are unchanged.
  - b. Will its quick ratio rise or fall?  
The quick ratio falls, since inventories are not included in the most liquid assets.

## Practice Problem

- If a company has a healthy current ratio but a significantly lower quick ratio, then you can assume that,
  - a. the cost of goods sold represents more than half of revenues.
  - b. current liabilities exceed current assets.
  - c. the firm sells only on a cash basis.
  - d. inventory represents a large portion of the firm's current assets.

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  - a. the cost of goods sold represents more than half of revenues.
  - b. current liabilities exceed current assets.
  - c. the firm sells only on a cash basis.
  - d. inventory represents a large portion of the firm's current assets.
- The current ratio's numerator (current assets) includes inventories, the quick ratio's numerator does not

## Practice Problem

- Which one of the following would be most detrimental to a firm's current ratio if that ratio is currently 2?
  - a. Collecting payment on an accounts receivable.
  - b. Selling marketable securities at cost.
  - c. Paying off accounts payable with cash.
  - d. Purchasing inventory on credit.

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  - b. Selling marketable securities at cost.
  - c. Paying off accounts payable with cash.
  - d. Purchasing inventory on credit.
- The next slides work through each option with an example



## Practice Problem

- We can think of an example,

|                              |          |                            |          |
|------------------------------|----------|----------------------------|----------|
| Cash + marketable securities | 0.5      | Debt due for repayment     | 0.5      |
| Accounts receivable          | 0.5      | Accounts payable           | 0.5      |
| Inventories                  | 1        |                            |          |
| <b>Current assets</b>        | <u>2</u> | <b>Current liabilities</b> | <u>1</u> |

## Practice Problem

- We can think of an example,

|                              |          |                            |          |
|------------------------------|----------|----------------------------|----------|
| Cash + marketable securities | 0.5      | Debt due for repayment     | 0.5      |
| Accounts receivable          | 0.5      | Accounts payable           | 0.5      |
| Inventories                  | 1        |                            |          |
| <b>Current assets</b>        | <u>2</u> | <b>Current liabilities</b> | <u>1</u> |

- $\text{Current ratio} = \text{Current assets} / \text{Current liabilities} = 2$

## Practice Problem

- Option a, cash rises and accounts receivable falls by the same amount, so current assets stay the same, and the current ratio stays the same

## Practice Problem

- Option a, cash rises and accounts receivable falls by the same amount, so current assets stay the same, and the current ratio stays the same
- Option b, some marketable securities turn into cash, but cash + marketable securities stays the same, so the current ratio stays the same

## Practice Problem

- Option c, suppose the firm pays down 0.5 of accounts payable, then

|                              |            |                            |            |
|------------------------------|------------|----------------------------|------------|
| Cash + marketable securities | 0          | Debt due for repayment     | 0.5        |
| Accounts receivable          | 0.5        | Accounts payable           | 0          |
| Inventories                  | 1          |                            |            |
| <b>Current assets</b>        | <b>1.5</b> | <b>Current liabilities</b> | <b>0.5</b> |

## Practice Problem

- Option c, suppose the firm pays down 0.5 of accounts payable, then

|                              |            |                            |            |
|------------------------------|------------|----------------------------|------------|
| Cash + marketable securities | 0          | Debt due for repayment     | 0.5        |
| Accounts receivable          | 0.5        | Accounts payable           | 0          |
| Inventories                  | 1          |                            |            |
| <b>Current assets</b>        | <b>1.5</b> | <b>Current liabilities</b> | <b>0.5</b> |

- Current ratio = Current assets / Current liabilities = 3

## Practice Problem

- Option d, suppose the firm acquires 1 of inventories on credit, then

|                              |          |                            |          |
|------------------------------|----------|----------------------------|----------|
| Cash + marketable securities | 0.5      | Debt due for repayment     | 0.5      |
| Accounts receivable          | 0.5      | Accounts payable           | 1.5      |
| Inventories                  | 2        |                            |          |
| <b>Current assets</b>        | <b>3</b> | <b>Current liabilities</b> | <b>2</b> |

## Practice Problem

- Option d, suppose the firm acquires 1 of inventories on credit, then

|                              |          |                            |          |
|------------------------------|----------|----------------------------|----------|
| Cash + marketable securities | 0.5      | Debt due for repayment     | 0.5      |
| Accounts receivable          | 0.5      | Accounts payable           | 1.5      |
| Inventories                  | 2        |                            |          |
| <b>Current assets</b>        | <b>3</b> | <b>Current liabilities</b> | <b>2</b> |

- Current ratio = Current assets / Current liabilities = 1.5



## Recap, Financing

- Financial leverage, debt raises shareholder returns in good times and lowers them in bad times
- Leverage ratios, how much debt the firm carries (debt ratios, times interest earned, cash coverage)
- Liquidity ratios, whether liquid assets can cover debt due soon (current and quick ratios)

# Key Takeaways

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- Shareholder value, the market-to-book ratio
- Investment, the returns on assets and equity, turnover ratios, and the profit margin
- Financing, financial leverage, leverage ratios, and liquidity ratios
- Corporate performance is multi-dimensional, no single ratio captures it

# Thank you!

Please let me know if you have any questions.